Orthogonal Invariance and Perturbative Stability of Weight-Space Offset Clustering

Abstract

We state and prove a structural invariance property of the spherical k-means clustering of weight-derived offset vectors in transformer language models, as used in the LARQL pipeline [Hayuk, 2026] and its downstream analyses [Author, 2026]. The main result is that if two model parameterizations differ by a joint orthogonal transformation of the residual-stream basis, then every quantity the offset-clustering methodology produces — anchoring functions, offsets, centroids, cluster partitions, per-cluster SVD spectra — is invariant up to covariant rotation. The pair partition induced by exact source-target token equality refines every such cluster partition. We then relax the exact-orthogonality hypothesis to an additive perturbation model. Offset angles deform linearly in the perturbation magnitude ε. Under an empirical-CDF regularity hypothesis on the cluster-margin distribution, the *global* cluster-assignment flip fraction is also linear in ε; per-cluster flip fractions can deviate from the global average, with empirical tail behavior that we characterize. Per-cluster, the centroid direction deflects by an angle scaling with the cluster's own flip fraction divided by its concentration, and the centroid-pair cosine deficit is quadratic in the same quantity. This separation identifies when averaged per-cluster statistics (centroid direction in cosine terms, d_{50}) are more stable than per-feature assignments across twin-trained models.

1 Setup and notation

Conventions. Throughout this note, $\|\cdot\|$ denotes the Euclidean (ℓ_2) norm on $\mathbb{R}^d$ and $\langle\cdot,\cdot\rangle$ the standard Euclidean inner product, so that $\|v\|^2 = \langle v, v\rangle = \sum_{i=1}^d v_i^2$. The unit sphere $S^{d-1} = \{v \in \mathbb{R}^d : \|v\| = 1\}$ is the unit sphere with respect to this norm. Matrix norms are never invoked; when we write $\|M\|$ for a matrix M it is understood to be applied row-wise to vectors in $\mathbb{R}^d$.

Fix a vocabulary V, a hidden dimension $d \in \mathbb{N}$, and a feed-forward intermediate dimension $d_{\text{ffn}} \in \mathbb{N}$. We consider transformer language models whose parameters relevant to the offset construction are the triple

$$\theta \;=\; (E,\, W_{\text{gate}},\, W_{\text{down}}) \;\in\; \mathbb{R}^{|V|\times d} \times \mathbb{R}^{d_{\text{ffn}}\times d} \times \mathbb{R}^{d_{\text{ffn}}\times d}.$$

Rows are indexed by vocabulary tokens for E and by feature indices for $W_{\text{gate}}, W_{\text{down}}$. For $t \in V$ we write $E[t] \in \mathbb{R}^d$ for the t-th row (the token embedding), and for a feature index $f \in \{1, \ldots, d_{\text{ffn}}\}$ we write $g_f^\theta = W_{\text{gate}}[f, :] \in \mathbb{R}^d$ for its gate vector and $w_f^\theta = W_{\text{down}}[f, :] \in \mathbb{R}^d$ for its down-projection vector. All three objects live in the d-dimensional residual-stream basis.

Remark 1.1. We adopt the convention that the down-projection vector w_f^θ is the row of W_{down} corresponding to feature f, so that both gate and down vectors are row vectors in $\mathbb{R}^d$ and transform identically under changes of residual-stream basis. This convention is consistent with the interpretation of FFN layers as key–value memories [Geva et al., 2021, 2022]: both the key (gate) and value (down) associated with feature f are vectors in the residual-stream representation space.

Let $V^{\text{word}} \subseteq V$ denote a fixed subset of tokens (in applications, the whole-word tokens). Let $\mathcal{F} \subseteq \{1, \ldots, d_{\text{ffn}}\}$ denote a fixed subset of features (in applications, those surviving a quality filter, a knowledge-band layer restriction, etc.).

Definition 1.2 (Anchoring)**.** The *source anchor* and *target anchor* functions induced by θ are $\sigma_\theta, \tau_\theta : \mathcal{F} \to V$ defined by

$$\sigma_\theta(f) \;=\; \operatorname*{arg\,max}_{t \in V^{\text{word}}} \cos\big(E[t],\, g_f^\theta\big), \qquad \tau_\theta(f) \;=\; \operatorname*{arg\,max}_{t \in V} \big\langle E[t],\, w_f^\theta \big\rangle,$$

where $\cos(x, y) = \langle x, y\rangle/(\|x\|\|y\|)$ and ties are broken by a fixed rule (e.g., smallest token index).

Definition 1.3 (Offset map)**.** Given θ and $f \in \mathcal{F}$ with $E[\tau_\theta(f)] \neq E[\sigma_\theta(f)]$, the *offset* of f in θ is the unit vector

$$\mathcal{O}(f;\theta) \;=\; \frac{E[\tau_\theta(f)] - E[\sigma_\theta(f)]}{\big\|E[\tau_\theta(f)] - E[\sigma_\theta(f)]\big\|} \;\in\; S^{d-1}.$$

For brevity we write $\mu_f^\theta = \|E[\tau_\theta(f)] - E[\sigma_\theta(f)]\|$ for the unnormalized offset magnitude.

Definition 1.4 (Spherical k-means)**.** Fix $K \in \mathbb{N}$. Given a set of unit vectors $\{x_f\}_{f \in \mathcal{F}} \subset S^{d-1}$ and a tuple of unit centroids $\mathbf{c} = (c_1, \ldots, c_K) \in (S^{d-1})^K$, the *assignment map* is

$$\pi(x; \mathbf{c}) \;=\; \operatorname*{arg\,max}_{k \in \{1,\ldots,K\}} \langle x, c_k\rangle,$$

with ties broken by a fixed rule. A *spherical k-means iteration* starting from $\mathbf{c}^{(t)}$ produces $\mathbf{c}^{(t+1)}$ via assignment and centroid update [Dhillon and Modha, 2001]:

(i) *Assignment:* let $S_k^{(t)} = \{f \in \mathcal{F} : \pi(x_f; \mathbf{c}^{(t)}) = k\}$.

(ii) *Update:* for each k, if $S_k^{(t)} \neq \emptyset$ and $\sum_{f \in S_k^{(t)}} x_f \neq 0$, set

$$c_k^{(t+1)} \;=\; \frac{\sum_{f \in S_k^{(t)}} x_f}{\big\|\sum_{f \in S_k^{(t)}} x_f\big\|};$$

otherwise, apply a fixed re-seeding rule (e.g., the point farthest from $c_k^{(t)}$ among currently assigned features).

2 The pair partition

Before addressing invariance under basis changes, we record a purely combinatorial fact about the offset construction that is independent of clustering.

Definition 2.1 (Pair partition)**.** Define an equivalence relation $\sim_P$ on $\mathcal{F}$ by

$$f \sim_P f' \quad\Longleftrightarrow\quad \big(\sigma_\theta(f), \tau_\theta(f)\big) = \big(\sigma_\theta(f'), \tau_\theta(f')\big).$$

Let $\mathcal{P}_\theta = \mathcal{F}/\sim_P$ denote the resulting partition, and for $[p] \in \mathcal{P}_\theta$ let $n_{[p]} = |[p]|$ be the *multiplicity* of the pair-class $[p]$.

Proposition 2.2 (Refinement)**.** *Let $\mathbf{c} \in (S^{d-1})^K$ be any centroid tuple, and let $\mathcal{C}(\mathbf{c}) = \{C_1, \ldots, C_K\}$ be the partition of $\mathcal{F}$ induced by the assignment map $f \mapsto \pi(\mathcal{O}(f;\theta); \mathbf{c})$. Then $\mathcal{P}_\theta$ refines $\mathcal{C}(\mathbf{c})$: every pair-class $[p] \in \mathcal{P}_\theta$ is contained in a single cluster C_k.*

Proof. For $f, f' \in [p]$, we have $\sigma_\theta(f) = \sigma_\theta(f')$ and $\tau_\theta(f) = \tau_\theta(f')$, so by Definition 1.3, $\mathcal{O}(f;\theta) = \mathcal{O}(f';\theta)$. The assignment $\pi(\cdot;\mathbf{c})$ is a function of the offset, hence $\pi(\mathcal{O}(f;\theta);\mathbf{c}) = \pi(\mathcal{O}(f';\theta);\mathbf{c})$ (ties notwithstanding, since the tie-breaking rule is deterministic). Thus f and f' lie in the same C_k. □

Remark 2.3. Proposition 2.2 holds for *any* deterministic clustering rule whose assignments depend only on the offset vector; the argument does not use any specific property of spherical k-means. The content-coherence diagnostic of [Author, 2026] is precisely the question of whether any pair-class of multiplicity ≥ 2 is contained in a given cluster.

3 Orthogonal invariance

Definition 3.1 (Orthogonal equivalence)**.** Two parameter tuples $\theta = (E, W_{\text{gate}}, W_{\text{down}})$ and $\theta' = (E', W'_{\text{gate}}, W'_{\text{down}})$ are *orthogonally equivalent*, written $\theta \sim_O \theta'$, if there exists $Q \in O(d)$ such that

$$E' \;=\; EQ, \qquad W'_{\text{gate}} \;=\; W_{\text{gate}}Q, \qquad W'_{\text{down}} \;=\; W_{\text{down}}Q.$$

Orthogonal equivalence is the natural gauge symmetry of the offset construction: it corresponds to a rotation of the residual-stream basis, which leaves all inner products between residual-stream representations invariant. Equivalence classes under $\sim_O$ are the true "observable" objects.

Lemma 3.2 (Anchoring invariance)**.** *If $\theta \sim_O \theta'$ via $Q \in O(d)$, then $\sigma_\theta = \sigma_{\theta'}$ and $\tau_\theta = \tau_{\theta'}$ as functions $\mathcal{F} \to V$.*

Proof. For any $t \in V$ and any $f \in \mathcal{F}$,

$$\cos\big(E'[t], g_f^{\theta'}\big) \;=\; \frac{\big\langle E[t]Q,\, g_f^\theta Q\big\rangle}{\|E[t]Q\|\,\|g_f^\theta Q\|} \;=\; \frac{\big\langle E[t], g_f^\theta\big\rangle}{\|E[t]\|\,\|g_f^\theta\|} \;=\; \cos\big(E[t], g_f^\theta\big),$$

using $\langle xQ, yQ\rangle = \langle x, y\rangle$ and $\|xQ\| = \|x\|$ for $Q \in O(d)$. Since the cosine similarity is preserved pointwise over t, the $\arg\max$ over t is the same for θ and θ', with identical tie-breaking. Hence $\sigma_{\theta'}(f) = \sigma_\theta(f)$. An identical argument, replacing cosine with inner product, shows $\tau_{\theta'}(f) = \tau_\theta(f)$. □

Theorem 3.3 (Orthogonal invariance of offset clustering)**.** *Let $\theta \sim_O \theta'$ via $Q \in O(d)$. Assume $E[\tau_\theta(f)] \neq E[\sigma_\theta(f)]$ for all $f \in \mathcal{F}$. Fix $K \in \mathbb{N}$ and an initial centroid tuple $\mathbf{c}^{(0)} = (c_1^{(0)}, \ldots, c_K^{(0)}) \in (S^{d-1})^K$. Define the corresponding initialization for θ' by $\mathbf{c}'^{(0)} = (c_1^{(0)}Q, \ldots, c_K^{(0)}Q)$. Let $\mathbf{c}^{(t)}$ and $\mathbf{c}'^{(t)}$ denote the centroids after t iterations of spherical k-means on offsets from θ and θ' respectively, using identical tie-breaking and re-seeding rules. Then for every $t \geq 0$:*

(i) $\mathcal{O}(f;\theta') = \mathcal{O}(f;\theta)\cdot Q$ *for every* $f \in \mathcal{F}$.

(ii) $c_k'^{(t)} = c_k^{(t)}Q$ *for every* $k \in \{1, \ldots, K\}$.

(iii) $\pi\big(\mathcal{O}(f;\theta');\mathbf{c}'^{(t)}\big) = \pi\big(\mathcal{O}(f;\theta);\mathbf{c}^{(t)}\big)$ *for every* $f \in \mathcal{F}$.

In particular, the spherical k-means cluster partition of $\mathcal{F}$ is identical for the two models at every iteration as a labeled set partition (under the correspondence $c_k \leftrightarrow c_kQ$), and the algorithms converge (if they do) in the same number of iterations.

Proof. Part (i). By Lemma 3.2, $\sigma_{\theta'} = \sigma_\theta$ and $\tau_{\theta'} = \tau_\theta$. Hence

$$\mathcal{O}(f;\theta') \;=\; \frac{E'[\tau_\theta(f)] - E'[\sigma_\theta(f)]}{\|E'[\tau_\theta(f)] - E'[\sigma_\theta(f)]\|} \;=\; \frac{(E[\tau_\theta(f)] - E[\sigma_\theta(f)])Q}{\|(E[\tau_\theta(f)] - E[\sigma_\theta(f)])Q\|} \;=\; \mathcal{O}(f;\theta)\cdot Q,$$

using $\|xQ\| = \|x\|$.

Parts (ii) and (iii) by induction on t.

Base case ($t = 0$). (ii) holds by construction. For (iii),

$$\pi\big(\mathcal{O}(f;\theta');\mathbf{c}'^{(0)}\big) \;=\; \arg\max_k \big\langle \mathcal{O}(f;\theta)Q, c_k^{(0)}Q\big\rangle \;=\; \arg\max_k \big\langle \mathcal{O}(f;\theta), c_k^{(0)}\big\rangle \;=\; \pi\big(\mathcal{O}(f;\theta);\mathbf{c}^{(0)}\big),$$

with identical tie-breaking since all inner products coincide.

Inductive step. Suppose (ii) and (iii) hold at iteration t. Let $S_k^{(t)} = \{f \in \mathcal{F} : \pi(\mathcal{O}(f;\theta);\mathbf{c}^{(t)}) = k\}$. By the inductive hypothesis (iii) at time t, the same set $S_k^{(t)}$ is the k-th cluster in θ'.

If $S_k^{(t)} \neq \emptyset$ and $\sum_{f\in S_k^{(t)}} \mathcal{O}(f;\theta) \neq 0$, then

$$c_k^{(t+1)} \;=\; \frac{\sum_{f\in S_k^{(t)}} \mathcal{O}(f;\theta)}{\big\|\sum_{f\in S_k^{(t)}} \mathcal{O}(f;\theta)\big\|}.$$

Using (i),

$$\sum_{f\in S_k^{(t)}} \mathcal{O}(f;\theta') \;=\; \sum_{f\in S_k^{(t)}} \mathcal{O}(f;\theta)\cdot Q \;=\; \Big(\sum_{f\in S_k^{(t)}} \mathcal{O}(f;\theta)\Big)Q,$$

which has norm $\big\|\sum_{f\in S_k^{(t)}} \mathcal{O}(f;\theta)\big\|$ because Q is orthogonal. Hence

$$c_k'^{(t+1)} \;=\; \frac{\big(\sum_{f\in S_k^{(t)}} \mathcal{O}(f;\theta)\big)Q}{\|\sum_{f\in S_k^{(t)}} \mathcal{O}(f;\theta)\|} \;=\; c_k^{(t+1)}\cdot Q,$$

establishing (ii) at time $t+1$. If the degenerate case of the update applies (empty cluster or zero vector sum), the fixed re-seeding rule is applied identically in both models by the inductive hypothesis, so the transformation relation is preserved.

For (iii) at time $t+1$:

$$\pi\big(\mathcal{O}(f;\theta');\mathbf{c}'^{(t+1)}\big) \;=\; \arg\max_k \big\langle \mathcal{O}(f;\theta)Q, c_k^{(t+1)}Q\big\rangle \;=\; \arg\max_k \big\langle \mathcal{O}(f;\theta), c_k^{(t+1)}\big\rangle \;=\; \pi\big(\mathcal{O}(f;\theta);\mathbf{c}^{(t+1)}\big).$$

The induction is complete. □

Corollary 3.4 (Invariance of per-cluster spectral statistics)**.** *Under the hypotheses of Theorem 3.3, for every $t \geq 0$ and every cluster index k, the singular value decomposition of the mean-centered offset matrix of cluster k is invariant. Specifically, if $X_k \in \mathbb{R}^{|S_k^{(t)}|\times d}$ is the matrix whose rows are $\{\mathcal{O}(f;\theta) - \bar{c}_k\}_{f\in S_k^{(t)}}$ (with $\bar{c}_k$ the unnormalized mean), and X_k' is the analogous matrix for θ', then $X_k' = X_k Q$. Consequently, X_k and X_k' have identical singular values; the right singular vectors of X_k' are those of X_k rotated by Q; and all invariant spectral statistics — including $d_{50}^{(k)}$, $d_{90}^{(k)}$, the spectral gap, and bootstrap confidence intervals — coincide for the two models.*

Proof. $X_k' = X_k Q$ follows from Theorem 3.3(i). The SVD of $X_k Q$ is $X_k Q = U_k \Sigma_k (V_k^\top Q) = U_k \Sigma_k (Q^\top V_k)^\top$, so the singular values coincide and right singular vectors rotate by Q. Functions of the singular value spectrum (including d_{50}) are therefore identical. □

4 Perturbative stability

We now relax the exact orthogonal equivalence of Definition 3.1 to an additive perturbation and study how the invariance of Theorem 3.3 degrades. This models the situation of two training runs with identical architecture and data but stochastic initialization, where we expect the two final parameterizations to be close to orthogonal equivalents of each other but not exactly equal.

Definition 4.1 (Perturbed orthogonal equivalence)**.** Fix $\theta = (E, W_{\text{gate}}, W_{\text{down}})$ and $Q \in O(d)$. Given perturbation matrices $\Delta_E \in \mathbb{R}^{|V|\times d}$, $\Delta_G, \Delta_D \in \mathbb{R}^{d_{\text{ffn}}\times d}$, the *perturbed parameterization* is

$$\theta' \;=\; \big(EQ + \Delta_E,\; W_{\text{gate}}Q + \Delta_G,\; W_{\text{down}}Q + \Delta_D\big).$$

Let $\varepsilon_E = \max_{t\in V} \|\Delta_E[t]\|$, $\varepsilon_G = \max_f \|\Delta_G[f,:]\|$, $\varepsilon_D = \max_f \|\Delta_D[f,:]\|$, and $\varepsilon = \max(\varepsilon_E, \varepsilon_G, \varepsilon_D)$.

We analyze how each component of the offset-clustering pipeline deforms under this perturbation. The technical foundation is the following elementary lemma, which controls the direction of a unit vector under additive perturbation.

Lemma 4.2 (Unit-vector deflection)**.** *Let $u \in S^{d-1}$, $c > 0$, and $v \in \mathbb{R}^d$. Write $v = v_\parallel u + v_\perp$, where $v_\parallel = \langle u, v\rangle \in \mathbb{R}$ is the scalar component of v along u and $v_\perp = v - v_\parallel u \in \mathbb{R}^d$ is the vector component of v orthogonal to u (so $\langle u, v_\perp\rangle = 0$). Suppose $c + v_\parallel > 0$. Then the vector*

$$u' \;=\; \frac{cu + v}{\|cu + v\|}$$

is well-defined and the angle α between u and u' satisfies

$$\tan\alpha \;=\; \frac{\|v_\perp\|}{c + v_\parallel}, \qquad \sin\alpha \;=\; \frac{\|v_\perp\|}{\sqrt{(c+v_\parallel)^2 + \|v_\perp\|^2}}, \qquad \cos\alpha \;=\; \frac{c + v_\parallel}{\sqrt{(c+v_\parallel)^2 + \|v_\perp\|^2}}.$$

In particular, for $\|v\| \ll c$,

$$\alpha \;=\; \frac{\|v_\perp\|}{c} + O\!\left(\frac{\|v\|^2}{c^2}\right), \qquad \cos\alpha \;=\; 1 \,-\, \frac{\|v_\perp\|^2}{2c^2} \,+\, O\!\left(\frac{\|v\|^3}{c^3}\right).$$

Proof. Write $cu + v = (c + v_\parallel)u + v_\perp$, so the components of $cu + v$ parallel and perpendicular to u are $c + v_\parallel$ and $\|v_\perp\|$ respectively. The claimed trigonometric identities follow from the definition of the angle between u and u' as $\cos\alpha = \langle u, u'\rangle$, using $\langle u, u'\rangle = (c + v_\parallel)/\|cu + v\|$ and $\|cu + v\| = \sqrt{(c+v_\parallel)^2 + \|v_\perp\|^2}$. The small-$\|v\|$ expansions are Taylor series of arctan and $1/\sqrt{1+x}$, respectively. □

Corollary 4.3 (Offset deflection)**.** *Let θ' be a perturbation of θ as in Definition 4.1. Fix $f \in \mathcal{F}$, and suppose $\sigma_{\theta'}(f) = \sigma_\theta(f)$ and $\tau_{\theta'}(f) = \tau_\theta(f)$ (i.e., the anchoring is unchanged; cf. Lemma 4.4 below). Let $\delta_f = \Delta_E[\tau_\theta(f)] - \Delta_E[\sigma_\theta(f)] \in \mathbb{R}^d$, and decompose $\delta_f = \delta_f^\parallel \cdot \mathcal{O}(f;\theta)Q + \delta_f^\perp$ with $\delta_f^\perp \perp \mathcal{O}(f;\theta)Q$. Then*

$$\mathcal{O}(f;\theta') \;=\; \frac{\mu_f^\theta \cdot \mathcal{O}(f;\theta)Q \,+\, \delta_f}{\|\mu_f^\theta \cdot \mathcal{O}(f;\theta)Q + \delta_f\|},$$

and the angle α_f between $\mathcal{O}(f;\theta)Q$ and $\mathcal{O}(f;\theta')$ satisfies

$$\alpha_f \;=\; \frac{\|\delta_f^\perp\|}{\mu_f^\theta} \,+\, O\!\left(\frac{\|\delta_f\|^2}{(\mu_f^\theta)^2}\right), \qquad \|\delta_f\| \le 2\varepsilon_E.$$

Proof. Since the anchoring tokens are unchanged, the numerator of $\mathcal{O}(f;\theta')$ is

$$E'[\tau_\theta(f)] - E'[\sigma_\theta(f)] \;=\; (E[\tau_\theta(f)] - E[\sigma_\theta(f)])Q + \delta_f \;=\; \mu_f^\theta \cdot \mathcal{O}(f;\theta)Q + \delta_f.$$

Apply Lemma 4.2 with $u = \mathcal{O}(f;\theta)Q$, $c = \mu_f^\theta$, $v = \delta_f$. □

Corollary 4.3 is the source of the linear dependence of the angular deflection on the perturbation: α_f scales as $\varepsilon_E/\mu_f^\theta$ to leading order. It is amplified for features whose raw offset magnitude μ_f^θ is small — i.e., features whose source and target embeddings lie close together in $\mathbb{R}^d$.

We next record an analogous result for anchoring stability. The situation is slightly messier because the anchoring arg max over the vocabulary is sensitive not only to the perturbation at the chosen token but also at competing tokens, and because cosine similarity involves normalization of *both* arguments.

Lemma 4.4 (Anchoring stability)**.** *Let θ' be a perturbation of θ as in Definition 4.1. Fix $f \in \mathcal{F}$ and define the* anchoring margins

$$\gamma_f^\sigma \;=\; \cos(E[\sigma_\theta(f)], g_f^\theta) \;-\; \max_{\substack{t\in V^{\text{word}}\\ t\neq\sigma_\theta(f)}} \cos(E[t], g_f^\theta), \qquad \gamma_f^\tau \;=\; \big\langle E[\tau_\theta(f)], w_f^\theta\big\rangle \;-\; \max_{\substack{t\in V\\ t\neq\tau_\theta(f)}} \big\langle E[t], w_f^\theta\big\rangle.$$

Let $\mu_E^{\min} = \min_{t\in V}\|E[t]\|$, assumed positive, and $M_E = \max_{t\in V}\|E[t]\|$. Assume $\|g_f^\theta\| > 0$ and $\|w_f^\theta\| > 0$. If

$$\frac{\varepsilon_E}{\mu_E^{\min}} + \frac{\varepsilon_G}{\|g_f^\theta\|} + R_\sigma \;<\; \frac{\gamma_f^\sigma}{2}, \tag{σ-cond}$$

$$M_E\,\varepsilon_D + \|w_f^\theta\|\,\varepsilon_E + \varepsilon_E\varepsilon_D \;<\; \frac{\gamma_f^\tau}{2}, \tag{τ-cond}$$

where $R_\sigma = O\big((\varepsilon_E/\mu_E^{\min})^2 + (\varepsilon_G/\|g_f^\theta\|)^2\big)$ captures second-order contributions, then $\sigma_{\theta'}(f) = \sigma_\theta(f)$ and $\tau_{\theta'}(f) = \tau_\theta(f)$.

Proof. We treat the two cases separately.

τ-case (inner product). For any $t \in V$, expand the perturbed inner product:

$$\begin{aligned}\big\langle E'[t], w_f^{\theta'}\big\rangle \;&=\; \big\langle E[t]Q + \Delta_E[t],\; w_f^\theta Q + \Delta_D[f,:]\big\rangle\\ &=\; \big\langle E[t], w_f^\theta\big\rangle \;+\; \underbrace{\big\langle E[t]Q, \Delta_D[f,:]\big\rangle}_{T_1(t)} \;+\; \underbrace{\big\langle \Delta_E[t], w_f^\theta Q\big\rangle}_{T_2(t)} \;+\; \underbrace{\big\langle \Delta_E[t], \Delta_D[f,:]\big\rangle}_{T_3(t)},\end{aligned}$$

where we used $\langle E[t]Q, w_f^\theta Q\rangle = \langle E[t], w_f^\theta\rangle$ since $Q \in O(d)$. By the Cauchy–Schwarz inequality and the perturbation bounds of Definition 4.1,

$$|T_1(t)| \;\le\; \|E[t]\|\cdot\varepsilon_D \;\le\; M_E\,\varepsilon_D, \qquad |T_2(t)| \;\le\; \varepsilon_E\cdot\|w_f^\theta\|, \qquad |T_3(t)| \;\le\; \varepsilon_E\,\varepsilon_D.$$

Therefore, uniformly over $t \in V$,

$$\Delta_\tau \;:=\; \max_{t\in V}\big|\langle E'[t], w_f^{\theta'}\rangle - \langle E[t], w_f^\theta\rangle\big| \;\le\; M_E\,\varepsilon_D + \|w_f^\theta\|\,\varepsilon_E + \varepsilon_E\,\varepsilon_D.$$

Under condition (τ-cond), $\Delta_\tau < \gamma_f^\tau/2$. For any $t \neq \tau_\theta(f)$,

$$\begin{aligned}\big\langle E'[\tau_\theta(f)], w_f^{\theta'}\big\rangle - \big\langle E'[t], w_f^{\theta'}\big\rangle \;&\ge\; \big(\langle E[\tau_\theta(f)], w_f^\theta\rangle - \Delta_\tau\big) - \big(\langle E[t], w_f^\theta\rangle + \Delta_\tau\big)\\ &\ge\; \gamma_f^\tau - 2\Delta_\tau \;>\; 0,\end{aligned}$$

so $\tau_\theta(f)$ is still the strict maximizer in θ' (the tie-breaking rule is not invoked). Hence $\tau_{\theta'}(f) = \tau_\theta(f)$.

σ-case (cosine similarity). We use an elementary bound on the perturbation of cosine similarity: for $x, y \in \mathbb{R}^d \setminus \{0\}$ and perturbations $\delta_x, \delta_y \in \mathbb{R}^d$ with $\|\delta_x\| < \|x\|$ and $\|\delta_y\| < \|y\|$,

$$\big|\cos(x+\delta_x, y+\delta_y) - \cos(x,y)\big| \;\le\; \frac{\|\delta_x\|}{\|x\|} + \frac{\|\delta_y\|}{\|y\|} + O\!\left(\frac{\|\delta_x\|^2}{\|x\|^2} + \frac{\|\delta_y\|^2}{\|y\|^2}\right). \tag{1}$$

To see (1), write $\hat{u} = u/\|u\|$ for $u \neq 0$. Then $\cos(x,y) = \langle \hat{x}, \hat{y}\rangle$, and by the bilinearity of the inner product,

$$\langle \widehat{x+\delta_x}, \widehat{y+\delta_y}\rangle - \langle \hat{x}, \hat{y}\rangle \;=\; \langle \widehat{x+\delta_x} - \hat{x}, \widehat{y+\delta_y}\rangle + \langle \hat{x}, \widehat{y+\delta_y} - \hat{y}\rangle.$$

Applying $|\langle a, b\rangle| \le \|a\|\|b\|$ and $\|\widehat{y+\delta_y}\| = \|\hat{x}\| = 1$,

$$\big|\langle \widehat{x+\delta_x}, \widehat{y+\delta_y}\rangle - \langle \hat{x}, \hat{y}\rangle\big| \;\le\; \|\widehat{x+\delta_x} - \hat{x}\| + \|\widehat{y+\delta_y} - \hat{y}\|.$$

By Lemma 4.2 applied with $u = \hat{x}$, $c = \|x\|$, $v = \delta_x$, the angle α_x between $\hat{x}$ and $\widehat{x+\delta_x}$ satisfies $\alpha_x = \|\delta_x^\perp\|/\|x\| + O(\|\delta_x\|^2/\|x\|^2)$, and therefore

$$\|\widehat{x+\delta_x} - \hat{x}\| \;=\; 2\sin(\alpha_x/2) \;\le\; \alpha_x \;\le\; \frac{\|\delta_x\|}{\|x\|} + O\!\left(\frac{\|\delta_x\|^2}{\|x\|^2}\right).$$

The analogous bound for $\widehat{y+\delta_y} - \hat{y}$ completes (1).

Now apply (1) with $x = E[t]Q$, $y = g_f^\theta Q$, $\delta_x = \Delta_E[t]$, $\delta_y = \Delta_G[f,:]$. Since $Q \in O(d)$ preserves norms, $\|x\| = \|E[t]\| \ge \mu_E^{\min}$ and $\|y\| = \|g_f^\theta\|$. Using $\cos(xQ, yQ) = \cos(x,y)$, we obtain, uniformly over $t \in V^{\text{word}}$,

$$\Delta_\sigma \;:=\; \max_{t \in V^{\text{word}}} \big|\cos(E'[t], g_f^{\theta'}) - \cos(E[t], g_f^\theta)\big| \;\le\; \frac{\varepsilon_E}{\mu_E^{\min}} + \frac{\varepsilon_G}{\|g_f^\theta\|} + R_\sigma,$$

where R_σ absorbs the second-order terms. Under condition (σ-cond), $\Delta_\sigma < \gamma_f^\sigma/2$, and the argmax-preservation argument of the τ-case, applied to cosine similarities instead of inner products, gives $\sigma_{\theta'}(f) = \sigma_\theta(f)$. □

Remark 4.5. Lemma 4.4 shows that the set of features whose anchoring flips under perturbation is supported on those with anchoring margin γ_f^σ or γ_f^τ less than $O(\varepsilon)$. The fraction of such features is, by definition, the empirical anchoring-margin CDF evaluated at an $O(\varepsilon)$ threshold; a quantitative bound on the expected flip fraction therefore requires a regularity assumption on the empirical CDF near zero (cf. Remark 4.8 for the corresponding assumption at the cluster-margin level). We do not claim a universal linear rate for anchoring flips; it is a conditional consequence of an assumption about the shape of the anchoring-margin distribution for a given model.

We now turn to cluster-assignment stability. For a feature whose anchoring is unchanged, the offset deflection is $\alpha_f \lesssim 2\varepsilon_E/\mu_f^\theta$ by Corollary 4.3. Whether this angular deflection induces a cluster-assignment flip depends on the feature's *cluster margin*, which we now define.

Definition 4.6 (Cluster margin)**.** Given centroids $\mathbf{c}$, feature $f \in \mathcal{F}$, and offset $x_f = \mathcal{O}(f;\theta) \in S^{d-1}$, let $k^*(f) = \pi(x_f; \mathbf{c})$ be the assigned cluster and $k^{**}(f) = \arg\max_{k \neq k^*(f)} \langle x_f, c_k\rangle$ the second-closest cluster. The *cluster margin* of f is

$$\gamma_f \;=\; \langle x_f, c_{k^*(f)}\rangle - \langle x_f, c_{k^{**}(f)}\rangle \;\in\; [0,2].$$

Lemma 4.7 (Sufficient condition for cluster-assignment stability)**.** *Let θ' be a perturbation of θ with anchoring unchanged at f, and let $\mathbf{c}, \mathbf{c}'$ be centroid tuples with $c'_k = c_k Q + \zeta_k$ for perturbations $\zeta_k \in \mathbb{R}^d$ with $\max_k \|\zeta_k\| \le \eta$ and $c'_k \in S^{d-1}$. If*

$$\alpha_f + \eta \;<\; \frac{\gamma_f}{2},$$

where α_f is the offset deflection angle from Corollary 4.3, then $\pi(\mathcal{O}(f;\theta');\mathbf{c}') = \pi(\mathcal{O}(f;\theta);\mathbf{c})$ under the correspondence $k \leftrightarrow k$.

Proof. For any k,

$$\big|\langle \mathcal{O}(f;\theta'), c'_k\rangle - \langle \mathcal{O}(f;\theta)Q, c_k Q\rangle\big| \;\le\; \alpha_f + \eta + O(\alpha_f \eta),$$

by Cauchy–Schwarz applied to the differences $\mathcal{O}(f;\theta') - \mathcal{O}(f;\theta)Q$ (of norm $\le \alpha_f$ to leading order, by Lemma 4.2) and $c'_k - c_k Q = \zeta_k$ (of norm $\le \eta$). Using $\langle \mathcal{O}(f;\theta)Q, c_k Q\rangle = \langle \mathcal{O}(f;\theta), c_k\rangle$, we conclude that each cosine similarity in θ' differs from its counterpart in θ by at most $\alpha_f + \eta$. If this is less than $\gamma_f/2$, then the largest cosine similarity is not displaced past the second-largest, and the argmax is preserved. □

Remark 4.8 (Assignment-flip fraction under an empirical-CDF regularity assumption)**.** Lemma 4.7 implies that a feature f flips its cluster assignment only if $\gamma_f < 2(\alpha_f + \eta) \lesssim 4\varepsilon_E/\mu_f^\theta + 2\eta$. Let F_N denote the empirical CDF of the margin distribution $\{\gamma_f\}_{f\in\mathcal{F}}$:

$$F_N(\tau) \;=\; \frac{1}{|\mathcal{F}|}\big|\{f \in \mathcal{F} : \gamma_f < \tau\}\big|.$$

Then the fraction of features that may flip their cluster assignment under the perturbation is bounded above by $F_N\big(2\max_f(\alpha_f + \eta)\big)$. This is an exact consequence of Lemma 4.7, but its smallness depends on the shape of F_N near zero.

To obtain a rate statement, we impose:

(R) There exist constants $L > 0$ and $\tau_0 > 0$ such that $F_N(\tau) \le L\tau$ for all $\tau \in [0, \tau_0]$.

Hypothesis (R) is a claim about the model and the clustering: it says that the empirical margin CDF grows at most linearly near zero. We do not prove (R); it is an empirically testable property of any given clustering, and different models or different choices of K may satisfy it with different constants L.

Under (R), for ε small enough that $2\max_f(\alpha_f + \eta) \le \tau_0$:

$$\text{fraction of flipped features} \;\le\; L \cdot 2\max_f(\alpha_f + \eta) \;=\; O(\varepsilon).$$

The constant depends on L, the distribution of offset denominators $\{\mu_f^\theta\}$, and η. The rate is linear in ε.

Remark 4.9 (Empirical verification of hypothesis (R) on Gemma-3 4B)**.** Hypothesis (R) is directly testable from any given clustering by computing the empirical margin distribution. We report the result of this test on one model: Gemma-3 4B with spherical k-means clustering ($K = 256$, $N = 18{,}324$ retained features, clustering procedure of [Author, 2026]).

Table 1 summarizes the empirical margin CDF.

Fitting the linear bound $F_N(\tau) \le L\tau$ on $[0, 0.1]$:

τ	0.01	0.05	0.10	0.15	0.33	0.44
$F_N(\tau)$	0.031	0.263	0.548	0.747	0.950	0.990

Table 1: Empirical margin CDF $F_N(\tau)$ for Gemma-3 4B, $K = 256$.

- least-squares slope through origin: $L_{\text{fit}} \approx 5.41$;
- worst-case bound: $L_{\text{worst}} \approx 11.05$.

Hypothesis (R) holds on this model with $\tau_0 = 0.1$ and $L = 11.05$. The worst-case constant is set by a small subpopulation of features with unusually small margins (3.1% of features have $\gamma_f < 0.01$). For the bulk of the population, the local slope $F_N(\tau)/\tau$ is closer to 5–6 in the range $\tau \in [0.01, 0.1]$.

The verdict for Proposition 4.10 under this value of L: by hypothesis (R), the *global* flip fraction p_{total} is bounded linearly in ε (with constant L); but the *per-cluster* flip fraction p_k that appears in Proposition 4.10's bound can be considerably larger than p_{total} at individual clusters. Empirically (see Remark 4.12), the distribution of p_k/p_{total} at this clustering has 95th-percentile ratio around 4–12 across tested noise scales; the maximum ratio can be 50 or more at very small σ. The centroid-stability bound therefore applies non-uniformly: for typical clusters (satisfying the bulk ratio), the cosine deficit is $O((L\varepsilon)^2/\rho_k^2)$ with constant coefficients of order 10^2–10^3 under typical $\rho_k \in [0.3, 0.7]$; for outlier clusters, the deficit is controlled by the cluster's own p_k, which may be a factor of 50 larger than p_{total}. The bound of Proposition 4.10 remains valid at all clusters; it is the *linearity in* ε that fails at outliers, because their p_k is not linear in ε at small ε.

We note that L depends on K (the number of clusters), the distribution of μ_f^θ, and the data-dependent structure of the offset cloud. The number reported here is specific to Gemma-3 4B at $K = 256$; a sweep over K would be informative but is not performed here.

Proposition 4.10 (Per-cluster centroid deflection and cosine deficit under perturbation)**.** *Fix $t \geq 0$ and let $\mathbf{c}^{(t)} = (c_1^{(t)}, \ldots, c_K^{(t)})$ be the centroid tuple at iteration t of spherical k-means on θ-offsets. Denote by $S_k^{(t)} \subseteq \mathcal{F}$ the cluster of features assigned to centroid k at iteration t, and let $n_k^{(t)} = |S_k^{(t)}|$, $s_k^{(t)} = \sum_{f \in S_k^{(t)}} \mathcal{O}(f;\theta)$, and $\rho_k^{(t)} = \|s_k^{(t)}\|/n_k^{(t)} \in (0,1]$ (the cluster concentration). Let $\mu_{\min}^\theta = \min_{f \in \mathcal{F}} \mu_f^\theta > 0$.*

Run spherical k-means on θ'-offsets with any initial centroid tuple $\mathbf{c}'^{(t)}$, producing iteration $t+1$ centroids $\mathbf{c}'^{(t+1)}$. Denote the iteration-$(t+1)$ clusters in θ' by $S_k'^{(t+1)}$ and define the per-cluster flip fraction

$$p_k \;=\; \frac{|S_k^{(t+1)} \triangle S_k'^{(t+1)}|}{n_k^{(t)}}$$

(the symmetric difference of the θ- and θ'-clusters that correspond to centroid k, normalized by the θ-cluster size). Assume:

(A1) For all $f \in \mathcal{F}$, $\sigma_{\theta'}(f) = \sigma_\theta(f)$ and $\tau_{\theta'}(f) = \tau_\theta(f)$ (anchoring preserved; sufficient conditions given by Lemma 4.4).

(A2) The previous-iteration centroids satisfy $\|c_k'^{(t)} - c_k^{(t)} Q\| \leq \eta^{(t)}$ for all k.

(A3) The quantity $\alpha_{\max} + p_k$ is small enough relative to $\rho_k^{(t)}$ that the linearization of Lemma 4.2 is valid at cluster k, i.e., the angle β_k defined below satisfies $\beta_k \leq \pi/4$. Here $\alpha_{\max} = 2\varepsilon_E/\mu_{\min}^\theta$ is the leading-order bound on offset deflection from Corollary 4.3.

Then

$$\left\|c_k'^{(t+1)} - c_k^{(t+1)}Q\right\| \;\le\; \sqrt{2}\,\beta_k, \qquad \beta_k \;\le\; \frac{2\alpha_{\max} + 2p_k}{\rho_k^{(t)}} \;+\; O\!\left(\frac{(\alpha_{\max} + p_k)^2}{(\rho_k^{(t)})^2}\right), \tag{2}$$

and the cosine deficit satisfies

$$1 - \cos\!\big(c_k^{(t+1)}Q,\, c_k'^{(t+1)}\big) \;=\; \frac{\beta_k^2}{2} + O(\beta_k^4) \;=\; O\!\left(\frac{(\alpha_{\max} + p_k)^2}{(\rho_k^{(t)})^2}\right). \tag{3}$$

Proof. Fix k. We drop the superscripts $^{(t)}$ and $^{(t+1)}$ where context permits, writing $S_k, n_k, s_k, c_k, \rho_k$ for the iteration-t quantities and c_k', c_k'' for iteration-$(t+1)$ centroids in θ and θ'.

Step 1: decompose the perturbed update.

By the spherical k-means update rule (Definition 1.4),

$$c_k^{(t+1)} = \frac{s_k}{\|s_k\|}, \qquad c_k'^{(t+1)} = \frac{s_k'}{\|s_k'\|}, \quad \text{where } s_k' = \sum_{f \in S_k'^{(t+1)}} \mathcal{O}(f;\theta').$$

Let $T_k = S_k^{(t)} \cap S_k'^{(t+1)}$ (features in cluster k at iteration t *and* at iteration $t+1$ in θ'), $P_k = S_k^{(t)} \setminus S_k'^{(t+1)}$ ("flipped out"), and $R_k = S_k'^{(t+1)} \setminus S_k^{(t)}$ ("flipped in"). Then $S_k'^{(t+1)} = T_k \cup R_k$ and $S_k^{(t)} = T_k \cup P_k$.

Under assumption (A1) and Corollary 4.3, for every $f \in \mathcal{F}$ we can write $\mathcal{O}(f;\theta') = \mathcal{O}(f;\theta)Q + \eta_f$ where $\|\eta_f\| \le \alpha_f + O(\alpha_f^2) \le \alpha_{\max} + O(\alpha_{\max}^2)$.

Therefore

$$\begin{aligned} s_k' &= \sum_{f \in T_k} \big(\mathcal{O}(f;\theta)Q + \eta_f\big) + \sum_{f \in R_k} \mathcal{O}(f;\theta') \\ &= \underbrace{\sum_{f \in S_k^{(t)}} \mathcal{O}(f;\theta)Q}_{=s_kQ} \;-\; \underbrace{\sum_{f \in P_k} \mathcal{O}(f;\theta)Q}_{A_k} \;+\; \underbrace{\sum_{f \in T_k} \eta_f}_{B_k} \;+\; \underbrace{\sum_{f \in R_k} \mathcal{O}(f;\theta')}_{C_k}. \end{aligned}$$

Step 2: bound the norms of A_k, B_k, C_k.

Each term in A_k and C_k is a unit vector; hence by the triangle inequality

$$\|A_k\| \le |P_k|, \qquad \|C_k\| \le |R_k|.$$

For B_k, the triangle inequality and Corollary 4.3 give

$$\|B_k\| \le \sum_{f \in T_k} \|\eta_f\| \le |T_k| \cdot \big(\alpha_{\max} + O(\alpha_{\max}^2)\big) \le n_k \cdot \alpha_{\max}\big(1 + O(\alpha_{\max})\big).$$

The churn count at cluster k is

$$m_k \;=\; |P_k| + |R_k| \;=\; |S_k^{(t+1)} \triangle S_k'^{(t+1)}| \;=\; p_k \cdot n_k^{(t)}$$

by the definition of p_k in the proposition statement.

Step 3: apply Lemma 4.2.

Set $u = c_k Q$, $c = \|s_k\|$, $v = -A_k + B_k + C_k$. Then $s'_k = s_k Q + v$ and $c_k'^{(t+1)} = (cu + v)/\|cu + v\|$. Decompose $v = v_\| u + v_\perp$ as in Lemma 4.2. By the triangle inequality,

$$\|v\| \;\le\; \|A_k\| + \|B_k\| + \|C_k\| \;\le\; m_k + n_k\alpha_{\max}\big(1 + O(\alpha_{\max})\big) \;=\; (p_k + \alpha_{\max})n_k + O(n_k\alpha_{\max}^2).$$

In particular, $\|v_\perp\| \le \|v\|$, so

$$\|v_\perp\| \;\le\; (\alpha_{\max} + p_k)\, n_k + O(n_k\alpha_{\max}^2). \tag{4}$$

By Lemma 4.2, the angle β_k between $u = c_k Q$ and $c_k'^{(t+1)}$ satisfies

$$\tan\beta_k \;=\; \frac{\|v_\perp\|}{\|s_k\| + v_\|}.$$

Since $|v_\|| \le \|v\| \ll \|s_k\|$ by assumption (A3) (which guarantees $\beta_k \le \pi/4$), we have $\|s_k\| + v_\| \ge \|s_k\|/2$. Combined with $\|s_k\| = \rho_k n_k$ and (4),

$$\begin{aligned}\beta_k \;\le\; \tan\beta_k \;\le\; \frac{\|v_\perp\|}{\|s_k\|/2} \;&=\; \frac{2\|v_\perp\|}{\rho_k n_k}\\ \le\; \frac{2(\alpha_{\max} + p_k)}{\rho_k} \;&+\; O\!\left(\frac{(\alpha_{\max} + p_k)^2}{\rho_k}\right).\end{aligned} \tag{5}$$

Under (A3), $\beta_k \le \pi/4 < 1$, so $\tan\beta_k \le 2\beta_k$; the factor 2 introduced here is absorbed into the leading constant. This establishes (2).

Step 4: Euclidean distance bound.

For unit vectors $x, y \in S^{d-1}$ with angle β between them, $\|x - y\| = 2\sin(\beta/2) \le \sqrt{2}\,\beta$ for $\beta \in [0, \pi/2]$. Applied to $x = c_k^{(t+1)}Q$ and $y = c_k'^{(t+1)}$ (noting $c_k^{(t+1)}Q$ is the normalization of $s_k Q$, which has the same direction as u), this gives the first inequality of (2).

Step 5: cosine deficit.

The cosine between unit vectors equals the cosine of the angle between them:

$$\cos\big(c_k^{(t+1)}Q,\, c_k'^{(t+1)}\big) \;=\; \cos\beta_k \;=\; 1 - \frac{\beta_k^2}{2} + O(\beta_k^4).$$

Substituting the bound (5), the cosine deficit is at most

$$\frac{1}{2\rho_k^2}\big(2\alpha_{\max} + 2p_k\big)^2 + O\!\left(\frac{(\alpha_{\max} + p_k)^3}{\rho_k^3}\right) \;=\; O\!\left(\frac{(\alpha_{\max} + p_k)^2}{\rho_k^2}\right).$$

This establishes (3). □

Remark 4.11 (Base case and initialization)**.** At $t = 0$, assumption (A2) holds with $\eta^{(0)} = 0$ if the iteration is initialized by $\mathbf{c}'^{(0)} = \mathbf{c}^{(0)}Q$. Proposition 4.10 then gives, for any cluster k,

$$\beta_k^{(1)} \;\le\; \frac{2(\alpha_{\max} + p_k^{(0,1)})}{\rho_k^{(0)}} \;+\; O\!\left(\frac{(\alpha_{\max} + p_k^{(0,1)})^2}{(\rho_k^{(0)})^2}\right),$$

where $p_k^{(0,1)}$ is the per-cluster flip fraction between the iteration-0 θ-clustering and the iteration-1 θ'-clustering. For $t \ge 1$ the assumption on $\eta^{(t)}$ must be supplied inductively, and p_k must be measured (or bounded) at each iteration; doing this rigorously requires tracking how $\rho_k^{(t)}$ and the per-cluster flip fractions evolve across iterations, which is beyond the scope of the present note.

Remark 4.12 (Role of hypothesis (R): bound on average p_k, not uniform p_k)**.** Hypothesis (R) of Remark 4.8 gives a bound on the *global* flip fraction $p_{\text{total}} \leq L \cdot 2\max_f(\alpha_f + \eta)$. Since $p_{\text{total}} = N^{-1}\sum_k n_k p_k$ is a size-weighted average of per-cluster flip fractions, (R) implies a bound on the *average* p_k under uniform cluster sizes, or on the weighted sum in general. However, (R) does *not* imply that every cluster satisfies $p_k \leq C \cdot p_{\text{total}}$ for any universal constant C: individual clusters may have p_k substantially larger than p_{total}.

Empirically, on the Gemma-3 4B spherical-k-means clustering of [Author, 2026] ($K = 256$, $N = 18{,}324$), we performed a synthetic perturbation experiment by adding iid Gaussian noise of scale σ to each offset, renormalizing, and reassigning. The results show that the ratio $\max_k p_k/p_{\text{total}}$ is substantial and *decreases* with σ: at $\sigma = 10^{-3}$ the maximum ratio is approximately 54, while at $\sigma = 10^{-2}$ it falls to approximately 8 and at $\sigma = 3 \times 10^{-2}$ to approximately 5. The 95th-percentile ratio across clusters is more stable, in the range 4–12 across all tested σ.

This has two consequences. First, the per-cluster statement of Proposition 4.10 is the correct formulation: attempting to bound every β_k in terms of p_{total} alone, via a uniform constant C, requires taking C much larger than the bulk-of-clusters value, and the resulting bound is loose for typical clusters while still not covering the tail. Second, the bound on β_k degrades gracefully for clusters with large p_k: the proposition continues to hold, but with the cosine deficit at those clusters scaling as p_k^2 rather than p_{total}^2. The tail of large-p_k clusters is a structural feature of the data — it corresponds to pairs of cluster centroids that are unusually close relative to their feature density — and deserves separate characterization (see Remark 4.13).

Remark 4.13 (Dependence on cluster concentration ρ_k)**.** The cosine-deficit bound $O(\varepsilon^2/\rho_k^2)$ deteriorates for clusters with low concentration. For a cluster whose members are spread uniformly over a hemisphere, $\rho_k = O(1/\sqrt{n_k})$ and the bound becomes $O(\varepsilon^2 n_k)$; for such clusters, centroid stability degrades with cluster size. For tight clusters with $\rho_k = \Omega(1)$, the bound is $O(\varepsilon^2)$ independent of n_k. The empirical cluster concentration ρ_k is directly measurable from the clustering output and can be reported alongside d_{50} and other per-cluster statistics.

5 Discussion

The results above formalize a rotation gauge symmetry of the weight-space offset-clustering pipeline. Theorem 3.3 establishes that every measurable output — anchoring functions, offset vectors, cluster centroids, cluster partitions, and by Corollary 3.4, per-cluster SVD spectra including d_{50} — is invariant under joint orthogonal reparameterization of the residual-stream basis. The invariance is exact, not merely up to permutation of cluster labels.

Proposition 2.2 records a distinct structural fact: the pair partition $\mathcal{P}_\theta$, defined by source-target token equality, refines *any* centroid-based cluster partition of the offset cloud. This implies that the content-coherence diagnostic of [Author, 2026] is well-defined independent of clustering: "coherent clusters contain a multiplicity-≥ 2 pair-class" is a statement about the refinement structure, not about the specific cluster assignment.

The perturbation analysis of Section 4 identifies three rates with distinct behaviors. The offset deflection angle α_f is linear in ε with amplification $1/\mu_f^\theta$ (Corollary 4.3). The cluster-assignment flip fraction is bounded by the empirical margin CDF evaluated at $O(\varepsilon)$ (Lemma 4.7); under hypothesis (R) of Remark 4.8 the *global* flip fraction p_{total} is linear in ε. The per-cluster flip fraction p_k, however, can deviate from p_{total} by cluster-dependent factors; empirically we observe max ratios of 50+ at small perturbation scales (Remark 4.12). Proposition 4.10 is therefore stated per-cluster: the centroid direction deflects by an angle $\beta_k = O((\alpha_{\max} + p_k)/\rho_k)$, and the centroid-pair cosine deficit $1 - \cos(c_k Q, c_k')$ is quadratic in $(\alpha_{\max} + p_k)$ because it equals $\sin^2(\beta_k/2) + O(\beta_k^4) = O(\beta_k^2)$.

The quadratic savings come from the geometric relationship between angle and cosine on the sphere, not from any cancellation between the two contributions to β_k (retained-feature offset-deflection and boundary churn), which are themselves both linear in their respective scales.

The qualitative upshot, contingent on hypothesis (R): *cluster geometry is more robust than cluster membership, on a square-root scale, for typical clusters.* Small perturbations to the parameter set produce changes in which specific features sit in which clusters at a globally linear rate in ε (global flip fraction is $O(L\varepsilon)$). For typical clusters, the per-cluster flip fraction p_k is of the same order as p_{total}; for such clusters, the centroid direction deflects by $O(L\varepsilon/\rho_k)$ and the cosine deficit is $O(L^2\varepsilon^2/\rho_k^2)$ — quadratically small in ε. For outlier clusters whose boundary with a neighbor is anomalously crowded, p_k can exceed p_{total} by large factors, and the centroid stability at those specific clusters degrades accordingly. Hypothesis (R) has been empirically verified on one model (Remark 4.9) with $L \approx 11$; the per-cluster flip-fraction ratio p_k/p_{total} has been characterized empirically on the same model (Remark 4.12). Averaged per-cluster statistics (centroid direction in cosine terms, cluster size, d_{50}) should be more stable across twin-trained models than per-feature assignments, with the quantitative strength of the stability bound varying across clusters.

Open questions

1. *Dependence of L in hypothesis (R) on K and model.* Remark 4.9 verifies (R) on one model (Gemma-3 4B at $K = 256$) with $L_{\text{worst}} \approx 11$. The quantitative strength of the global flip-fraction bound of Remark 4.8 depends directly on L, and L is expected to vary with K (more clusters crowd decision boundaries, inflating L) and with the model (different offset-cloud structure). A sweep of L as a function of K for a fixed model, and across several models, would sharpen the practical applicability of the bounds. An open conjecture is that coherent clusters in the sense of [Author, 2026] have systematically larger margins than noise clusters; if so, restricting the margin CDF to coherent features would give a smaller L, and the stability claims would apply more strongly to the methodologically relevant sub-population.

2. *Characterizing the per-cluster flip distribution.* Remark 4.12 documents that the ratio $\max_k p_k/p_{\text{total}}$ depends strongly on the perturbation scale: approximately 54 at $\sigma = 10^{-3}$, falling to approximately 5 at $\sigma \geq 3 \times 10^{-2}$. The small-$\sigma$ regime is where training-stochasticity modeling lives, and it is precisely where the ratio is largest. A structural characterization of which clusters incur high p_k at small σ — e.g., whether they are the clusters with anomalously close centroid neighbors, or the clusters with content-coherence properties correlated with boundary crowding — would refine Proposition 4.10's applicability across the cluster population.

3. *Cross-model case.* Theorem 3.3 addresses twin models whose parameterizations differ by joint residual-stream rotation. Genuine cross-model comparison (e.g., Gemma vs. OLMo-2) requires a map between two different embedding spaces, with different vocabularies and dimensionalities. A natural candidate is Procrustes alignment of shared-vocabulary embeddings, but its effect on the downstream clustering is not characterized by the results here.

4. *Exact centroid-perturbation bounds via perturbation theory.* Proposition 4.10 bounds β_k and cosine deficit via direct argument using Lemma 4.2 and a decomposition of the perturbed update into retained-feature and churn contributions. An alternative rigorous approach would apply singular-subspace perturbation theory — Davis–Kahan [Davis and Kahan, 1970] or Wedin [Wedin, 1972] — to the sample-covariance operator of the offset cloud, yielding bounds that are tighter for tightly concentrated clusters at the cost of stronger hypotheses on the spectrum. The corresponding result for eigenvalue stability (d_{50}-deformation) under small perturbations is a natural target

and would close the gap between our Corollary 3.4 (exact invariance under the orthogonal case) and a robust version under perturbation.

5. *Random-perturbation regime.* If $\Delta_E, \Delta_G, \Delta_D$ are modeled as random matrices (e.g., iid Gaussian entries, as a model for training-stochasticity residuals), concentration estimates can sharpen the global flip-fraction bound into a quantitative probability statement. This requires integrating Lemma 4.7 against the perturbation distribution and combining with hypothesis (R) or its failure modes. Beyond the global bound, characterizing the distribution of p_k under random perturbation — in particular its concentration and heavy-tail behavior — would complete the robustness picture.

6. *Connection to pair-partition stability.* The pair partition $\mathcal{P}_\theta$ is determined by $\sigma_\theta, \tau_\theta$. By Lemma 4.4, anchoring is stable up to features in the $O(\varepsilon)$ tail of the anchoring-margin distribution. Consequently, $\mathcal{P}_{\theta'}$ differs from $\mathcal{P}_\theta$ only by a small number of reassignments at the boundary, under an analogue of hypothesis (R) for the anchoring margin. A clean statement of pair-partition stability as a set-partition distance (e.g., variation of information, Rand index) is not treated here but follows the same pattern.

Acknowledgments. This note builds on the offset construction of [Hayuk, 2026] and the methodology analysis of [Author, 2026]. Spherical k-means is due to [Dhillon and Modha, 2001]; the perturbation-theoretic framework draws on classical results of [Davis and Kahan, 1970, Wedin, 1972, Stewart and Sun, 1990].

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